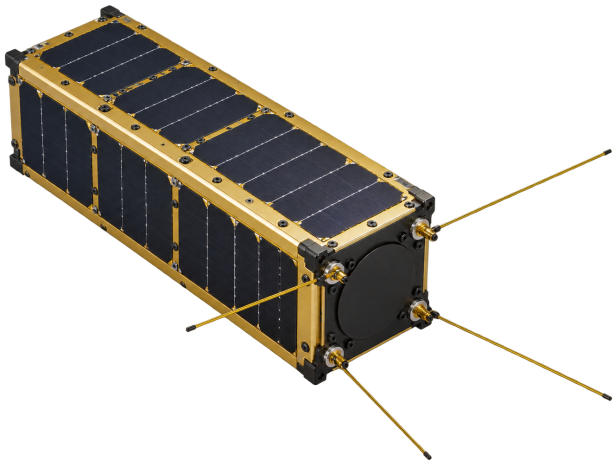
SMART-QSO

A small spacecraft. A wider conversation.

An open 3U packet relay with bounded autonomy and an isolated onboard language experiment.



Concept illustration. No flight unit, NASA affiliation, qualification or secured launch is implied.

Funding target: $500k-$1.5m. Initial feasibility milestone: $75k.

Prepared for educational/nonprofit sponsors, amateur-radio partners and technology-research funders. All cost allocations and hardware budgets remain subject to quotations and measured closure.

A useful service with a testable research case

SMART-QSO stores short public messages from licensed amateur stations and relays them to recipients during later contacts. It starts with ordinary AX.25/1200-baud AFSK and KISS-compatible station equipment. This is a pilot amateur service and an open research platform, not a commercial or emergency-critical communications promise.

The deterministic MCU controls packet capacity, fairness, retries, power limits and radio inhibits. Small predictors may recommend resource use within those limits. The separate language computer can explain verified health but cannot command the spacecraft or rewrite user messages.

Service level	Measured success target
Minimum	30 days of commissioning, verified telecommand, safe fallback and independent-station delivery
Full	1,000 corroborated acknowledged messages; 20 stations; 3 regions; 90 days
Research	Matched delivered-message/Wh comparison, uncertainty analysis and zero safety-limit violations
Extended	12 months of useful operations and reproducible engineering results

What makes the program worth supporting

A compatible public service gives the mission an observable purpose. The autonomy experiment has a conventional baseline and charges compute energy to its own benefit claim. Open protocols, source code, engineering evidence and public results let schools and clubs participate and let reviewers distinguish software demonstrations from flight readiness.

Existing packet satellites and onboard-AI experiments are part of the comparison, not achievements to erase with a first-of-its-kind claim. Community adoption, measured energy benefit and rigorous failure containment are the research questions.

Safety has a separate owner

MCU / essential service	Optional language payload
Safety limits, watchdog and persistent state	Read-only verified health snapshots
Relay scheduling and public message integrity	Constrained status explanation only
Factual telemetry and template beacons	Deadline, freshness, ASCII and factuality validation
Signed telecommands and retained RF shutdown	Power-gated; failure returns to templates

Command and update assurance

Ed25519 authentication is distinct from readable radio content. The shared C core rejects invalid signatures and replay, commits the replay floor before accepting control, and fails closed on storage faults. Signed firmware manifests bind version, length and content. Reference boot trials and rollback are tested on the host; physical write protection, power-cut behavior and board acceptance remain open.

Reference firmware enforces SAFE/RF OFF when verified EPS input is unavailable. Hardware execution remains unverified; a compiling ARM image does not demonstrate a working spacecraft interface.

NASA-aligned process

NPR 7150.2D and NASA-STD-8739.8B supply software-engineering, assurance and safety tailoring inputs. Mission risk class, software classification and safety criticality require separate assessment. Requirement-level evidence, independent review and accepted tailoring are necessary before any compliance declaration.

3U feasibility is the first physical gate

The mass, volume, energy, flash, RF and data budgets are executable, share one source of assumptions and explicitly report unclosed feasibility. Estimated positive margins are not measured hardware results.

Budget	Evidence required before closure
Power / current	Whole cold-boot/model-load/inference/shutdown cycle; RF PA efficiency; inrush; battery cycling and EOL
Thermal / radiation	Conductive vacuum cooling, correlated hot/cold model, TVAC and selected-parts TID/SEE/latch-up assessment
Mechanical / volume	Supplier CAD, deployer ICD, tolerance stacks, mass/CG/inertia, structural and mechanism tests
RF / service	Measured sensitivity/gain/emissions, modem interoperability, Doppler/desense, realistic contact exposure
Disposal / safety	Accepted probabilistic lifetime and reentry/collision analysis; independent passivation and launch inhibits

Design boundaries

Keep the 3U reference fixed until a documented scope decision. Prefer selected heritage bus components and make custom design sources, exact parts, harness interfaces and manufacturing inspection records explicit. The parametric packaging file is a concept envelope, not a fabrication package.

The amateur space-station control path must demonstrate telecommanded cessation. Current 47 CFR 97.207 includes a five-year post-mission disposal criterion for applicable LEO spacecraft. Orbit selection and uncertainty analysis must support that requirement; a blanket historical 25-year assumption is superseded.

A program advanced by evidence

Gate	Required evidence
Resurrection baseline	controlled mission definition; complete artifact inventory; reproducible host software demo; audit and actionable findings
Feasibility / SRR	closed preliminary 3U budgets; measured AI energy/thermal profiles; regulatory pathway; staffing and supplier estimates
PDR	selected hardware and ICDs; credible packaging; flatsat evidence; funded engineering model
CDR	fabrication package; target/HIL interfaces; hazard controls verified; review actions accepted
TRR / environmental	engineering model; calibrated facilities; correlated budgets; controlled executable procedures
FRR / ORR	flight unit acceptance; hazard closure; regulatory approval; operators rehearsed; provider acceptance
On-orbit / closure	commissioning logs; acknowledged relay deliveries; matched research results; verified passivation and disposal reporting

Readiness planning envelope: 24-36 months after staffing and funding assumptions are validated. This is not a promised launch date. Qualification facilities, supplier lead times, institutional reviews and launch opportunities determine the critical path.

The team is not yet staffed. Named accountable owners are required for program management, systems, RF/licensed operations, EPS/avionics, mechanics/thermal/radiation, flight software, payload/ground, assurance and integration/operations.

Funding allocations and the first ask

Illustrative allocations only. Base cash: $890,000. Cash reserve (25%): $222,500. Total cash scenario: $1,112,500. Uncommitted in-kind contributions: $205,000. Neither quotes nor partner commitments are secured.

Category	Cash allocation	In-kind, uncommitted
Engineering labor	$350,000	$120,000
Bus, payload, spares and hardware	$180,000	$0
Ground stations and GSE	$20,000	$10,000
Environmental and integration facilities	$90,000	$30,000
Launch and integration	$120,000	$0
Mission operations	$60,000	$30,000
Independent assurance and regulatory work	$70,000	$15,000
Cash reserve	$222,500	Not assigned

Initial feasibility milestone: $75,000

Deliver calibrated payload-cycle energy/thermal profiles; real station/TNC interoperability; preliminary 3U budgets with uncertainties; a sponsoring institution and named subsystem owners; independent requirements review; and a supplier quote log.

NASA CubeSat Launch Initiative may provide a launch-support route for eligible educational/nonprofit organizations. It is not an award, a launch commitment or spacecraft-development funding in this proposal. Scope, eligibility and cost must be reviewed against the actual opportunity.

Evidence, sources and open work

The repository contains the complete audit inventory and finding register, controlled baseline, requirements, work packages, standards tailoring matrix, software tests, reference firmware, ground demo, verification campaign and research protocol. Historical reviews remain accessible but explicitly unverified.

Outstanding physical work includes selected hardware/driver integration, fabrication sources, flatsat and engineering model, calibrated RF/environmental/radiation testing, hazard closure, named independent reviews, licensing/coordination, launch acceptance and commissioning.

Source	Reference
Software engineering	nodis3.gsfc.nasa.gov - NPR 7150.2D
Software assurance / safety	standards.nasa.gov - NASA-STD-8739.8B
Environmental guidance	standards.nasa.gov - GSFC-STD-7000B (GEVS)
Amateur space station control / disposal	ecfr.gov - 47 CFR 97.207
Amateur permissible service	ecfr.gov - 47 CFR 97.113
Launch support eligibility	nasa.gov/kennedy/launch-services-program/cubesat-launch-initiative/
Program and engineering evidence	mission/current/; docs/resurrection/; output/verification/

Next decision

Fund the measurable feasibility gate, establish institutional ownership and review the resulting evidence before committing to spacecraft fabrication. The value of the project is a useful service and an honest experiment whose safety, benefit and maturity can be inspected.